| INCH-POUND |

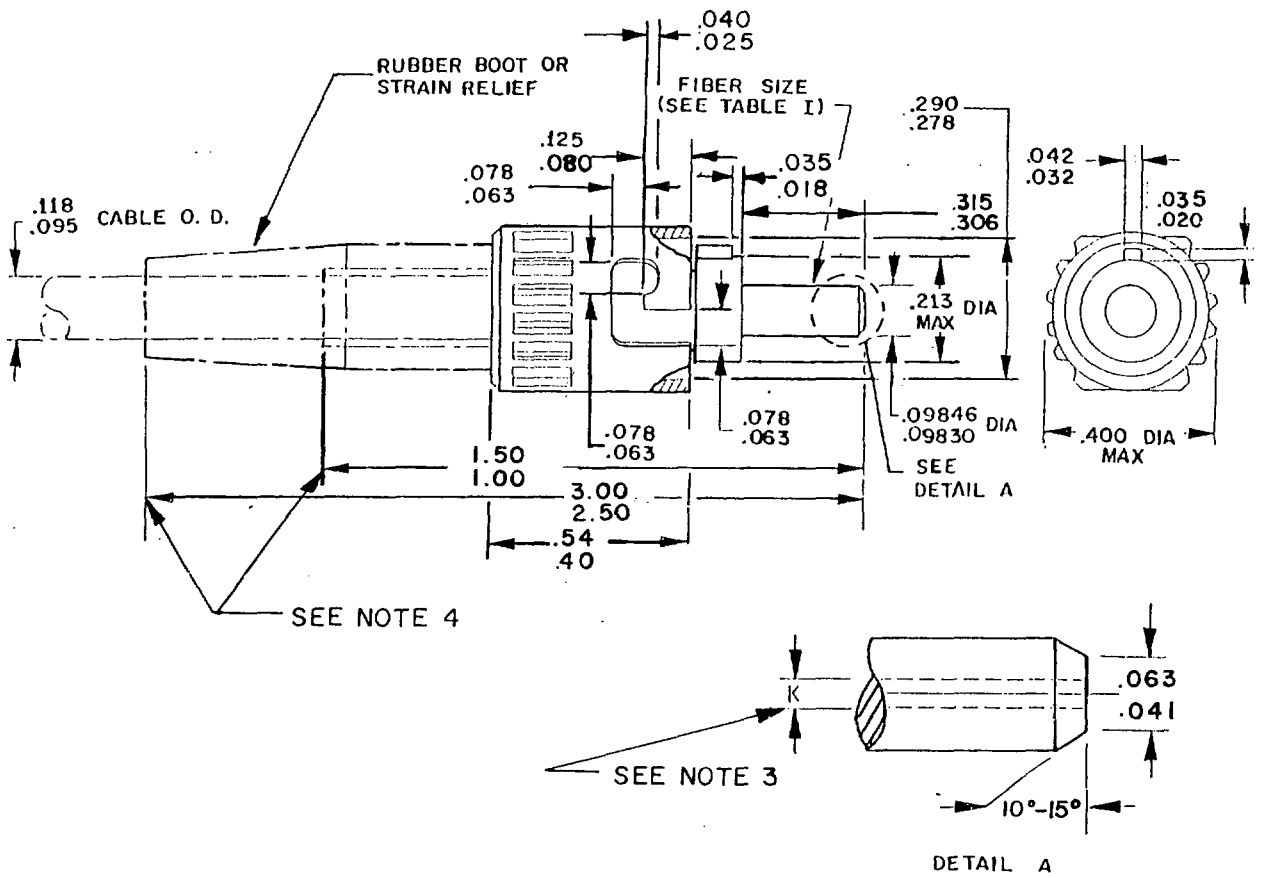
MIL-C-83522/12A
25 January 1990
SUPERSEDING
MIL-C-83522/12
21 October 1988

MILITARY SPECIFICATION SHEET

CONNECTOR, FIBER OPTIC, PLUG-RECEPTACLE-ADAPTER STYLE, 2.5 MM BAYONET CONNECTOR, PLUG, EPOXY, FOR 50/125 µm, 62.5/125 µm and 100/140 µm

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83522.



(A) FIGURE 1. Dimensions and configuration.

(A) denotes changes

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Inches	mm	Inches	mm	Inches	mm
.018	0.46	.095	2.41	.315	8.00
.020	0.51	.09830	2.4968	.400	10.16
.025	0.64	.09846	3.5009	.40	10.2
.032	0.81	.118	3.00	.54	13.7
.035	0.89	.125	3.18	1.00	25.4
.040	1.02	.213	5.41	1.25	31.8
.042	1.07	.278	7.06	2.50	63.50
.063	1.60	.290	7.36	3.00	76.2
.078	1.98	.306	7.77		
.080	2.03	.314	7.98		

NOTES:
1. Dimensions are in inches.
2. Metric equvalents are given for general information only.
3. K dimension is specified in µm. This dimension is critical to fiber alignment. Dimensions for fiber optic fiber shall be in accordance with DOD-F-49291.
4. Dimension indicates assembled configuration.

Ⓐ FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Metals: The plug housing shall be of corrosion-resistant steel in accordance with QQ-S-763, class 303, /316; nickel-plated brass in accordance with QQ-B-626; nickel-plated zinc die cast alloy; or 6061 nickel-plated aluminum. Other metal components shall be of corrosion-resistant steel in accordance with QQ-S-763, class 303, /316; nickel-plated beryllium copper in accordance with QQ-B-530; nickel-plated brass in accordance with QQ-B-626, or 6061 nickel-plated aluminum.

(A) Epoxies: Use Trabond 230, Eccobond 144B, Hysol 0151, or an equivalent epoxy approved by the qualifying activity.

Dimensions and configuration: See figure 1.

Fiber optic cable requirements:

Cable diameter: In accordance with DOD-C-85045/2.

Cable configuration: In accordance with DOD-C-85045/2.

Fiber diameter: In accordance with DOD-F-49291/1 through DOD-F-49291/4.

Fiber optic contact:

Method of optical alignment: Sleeve/ferrule (bayonet).

Lens configuration: Not applicable.

Coating requirements: Not applicable.

Optical requirements:

Number of optical terminus: One.

Coupling loss (attenuation): 1 dB maximum per mated pair.

Weight: 4 grams maximum.

Polarization: Key/slot (bayonet).

Safety wire holes: Not required.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: Not applicable.

Coupling proof torque: Not applicable.

Coupling mechanism retention force: The force shall be 20 pounds minimum and shall not result in any physical damage.

Mating durability: In accordance with MIL-C-83522, mating durability paragraphs, except 500 cycles instead of 200 cycles.

Tensile load: In accordance with MIL-C-83522, tensile load paragraphs, except 9 kilograms instead of 18 kilograms.

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Cable retention: The load shall be 50 pounds minimum and shall not result in any physical damage. Optical properties shall be monitored before and after load. The load shall not be applied to the coupling mechanism.

Optical decoupling: The force required to optically decouple the connector shall be one pound minimum.

Submersion: Not applicable.

Ozone exposure: Not applicable.

A Military Part or Identifying Numbers (PINs): See table I.

A TABLE I. Part or Identifying Numbers (PINs).

PIN M83522/12-	Fiber size	K dimension (µm)	
		GO 1/	NO-GO 2/
01	50/125, 62.5/125	127	132
02	100/140	143	148
03	50/125, 62.5/125 3/	127	132
04	100/140 3/	143	148

1/ GO designates the smallest precision gauge pin the hole must accept. This gauge pin should have a tolerance of +0.5, -0 microns.
2/ NO-GO designates the smallest precision gauge pin the hole must not accept. This gauge pin should have a tolerance at +.5, -0 microns.
3/ Salt spray, fluid immersion not applicable.

A Plug accessories: Each plug shall be supplied with a minimum of one crimp ferrule; one protective cable-end cap; one 1.5 inch length of heat-shrink tubing or one boot; one set of instructions; and listing of the epoxies specified in this specification sheet.

Note: The bayonet connector mates with MIL-C-83522/13 adapter bayonet and MIL-C-83522/14 PC mount.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0082-1)

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